

CODE REVIEW PLATFORM

MINOR PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this minor project report for the course **21CSC203P ADVANCED PROGRAMMING PRACTICE** entitled in "**CODE REVIEW PLATFORM**" is the bonafide work of **Aaron Varughese Binu (RA22110030010171)**, **Joel Jacob (RA22110030010182)** who carried out the work under my supervision.

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ABSTRACT

The Code Review Platform Project introduces a comprehensive solution for optimizing the code review process within software development teams. Recognizing the pivotal role of code reviews in ensuring code quality, bug detection, and fostering collaboration among team members, our platform is designed to streamline and enhance this essential aspect of the software development lifecycle. With an intuitive web interface, developers can seamlessly submit, manage, and discuss code changes in real-time. The platform integrates seamlessly with popular version control systems such as Git and SVN, ensuring efficient synchronization and automated tracking of code changes. A key emphasis is placed on fostering collaboration, featuring threaded discussions and precise commenting features for detailed feedback. Additionally, the platform incorporates automated code quality metrics and checks, supporting static code analysis tools to provide valuable insights into coding standards and best practices. By implementing role-based access control and customizable workflows, the Code Review Platform adapts to diverse team structures and development processes. Ultimately, this project endeavors to significantly improve the efficiency of code reviews, resulting in heightened code quality, reduced bugs, and increased collaboration, thereby contributing to the overall success of software projects

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CHAPTER 1 - INTRODUCTION

The introduction of the Code Review Platform Project marks a significant stride in the realm of software development methodologies. Code review, a critical practice in the software development lifecycle, plays a vital role in ensuring code quality, identifying potential issues, and promoting collaborative efforts within development teams. This project endeavors to create an innovative and efficient solution that addresses the challenges associated with traditional code review processes. By providing a centralized web-based platform, the project aims to enhance the overall experience of submitting, reviewing, and discussing code changes. With a focus on collaboration, integration with version control systems, and the incorporation of automated code quality checks, the Code Review Platform aims to revolutionize the way development teams approach and execute code reviews. This introduction sets the stage for a detailed exploration of the platform's features, implementation, and the anticipated impact on code quality and team collaboration in software development projects.

CHAPTER 2 - LITERATURE SURVEY

Code Review Platform Training, Testing, and Accreditation

1. Variability in Testing and Accreditation: The literature on variability in testing and accreditation highlights the dynamic nature of software development and the challenges associated with ensuring consistent quality across diverse systems. Research in this domain delves into the complexities of testing methodologies tailored to handle the variability inherent in different software environments. Variability may arise from factors such as diverse user requirements, changing business needs, and evolving technology stacks. Studies explore innovative testing approaches, including model-based testing, mutation testing, and adaptive testing strategies, to address variability effectively. Additionally, the literature emphasizes the importance of accreditation frameworks in validating testing processes, ensuring adherence to industry standards, and ultimately certifying the reliability of software systems.

2. Challenges in Maintaining Quality: The body of literature addressing challenges in maintaining quality in software development provides insights into the multifaceted issues faced by development teams. Research investigates factors contributing to challenges in maintaining quality, including the rapid pace of technology evolution, increasing system complexity, and the demand for frequent releases. Studies delve into the impacts of shifting requirements, resource constraints, and the need for agile methodologies on quality assurance processes. Furthermore, the literature explores the role of automated testing, continuous integration, and effective collaboration within development teams as strategies to overcome challenges and ensure sustained software quality.

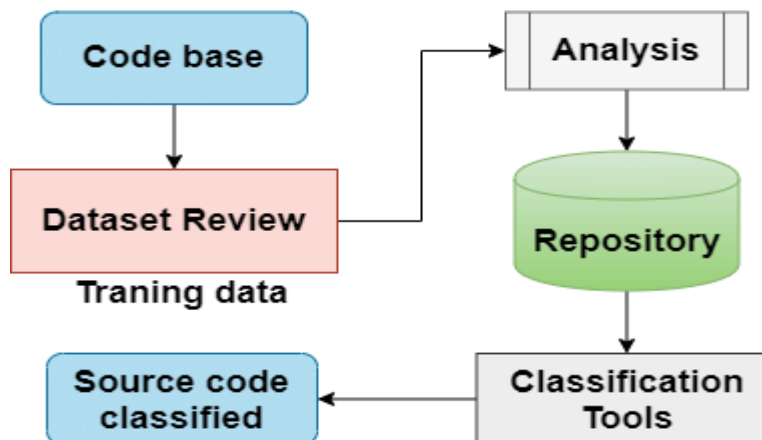
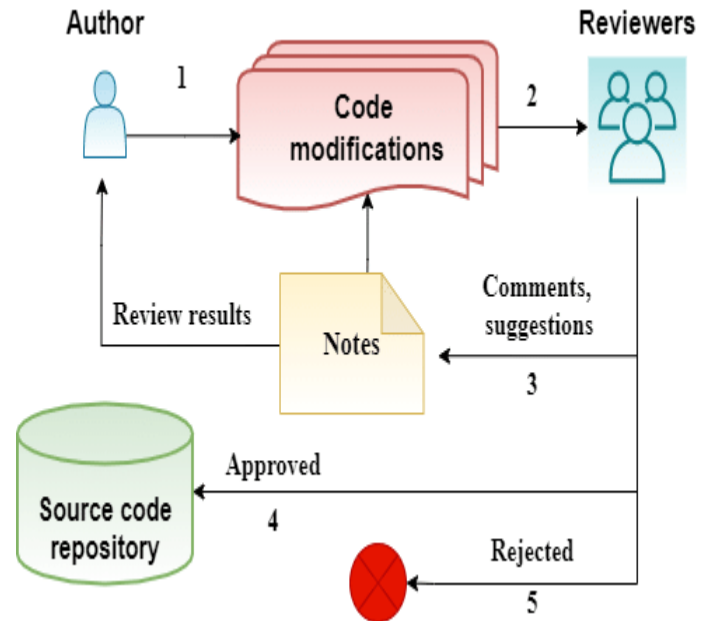
3. Global Significance of Interpreter Training: Research in this area focuses on the challenges and opportunities in interpreter training programs, emphasizing the need for standardized practices to maintain a high level of proficiency. Studies discuss the impact of interpreter training on the quality of language services in various contexts, including legal, healthcare, and diplomatic settings.

The Code Review Platform is a sophisticated software solution designed to streamline and enhance the code review process within software development teams. With a focus on improving code quality and developer collaboration, this platform goes beyond traditional code review tools by incorporating a unique capability: the automatic analysis and correction of basic syntax errors. Developers can submit their code to the platform, and the system, equipped with advanced algorithms, rigorously examines the codebase for common syntax pitfalls such as missing semicolons, braces, and quotation marks. Upon detection, the platform not only highlights these errors but also provides an immediate and accurate correction, allowing developers to receive feedback and corrected code promptly.

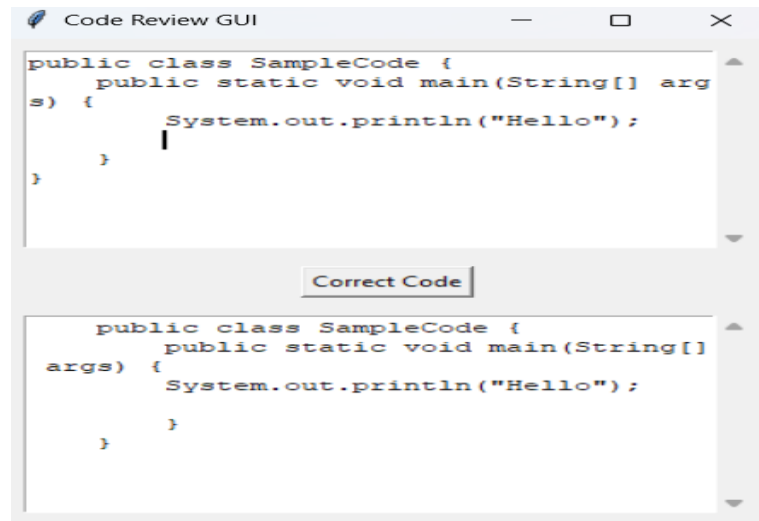
This innovative feature significantly accelerates the code review process, enabling developers to address syntax errors efficiently and adhere to coding standards seamlessly. By automating the identification and correction of basic syntax errors, the Code Review Platform promotes a more efficient development workflow, reduces the likelihood of introducing bugs related to syntax issues, and ultimately contributes to elevated code quality. This platform emerges as a valuable asset in modern software development, offering a blend of collaborative code review and automated error correction to foster a more productive and error-resistant coding environment.

CHAPTER 3 – SYSTEM DESIGN

4.1 Block Diagram



4.2 Design



4.3 Module description

The system design of the code review platform comprises several interconnected modules. The Version Control System Integration Module synchronizes with Git repositories, and the Notification Module keeps users informed. Code Analysis tools in a dedicated module identify potential issues, and the Reporting and Analytics Module offers insights. Optional integrations with CI/CD tools enhance development workflows. . Together, these modules create a comprehensive and adaptable system supporting efficient and collaborative code reviews.

CHAPTER 4 – REQUIREMENT ANALYSIS

3.1 Hardware Requirement

1. Server:

- A server to host the code review platform.

2. Database Server:

- If the platform uses a database (which is likely), a separate server or dedicated resources for the database. The database server's specifications will depend on the volume of data and concurrent database transactions.

3.2 Software Requirement

1. Operating System:

- The choice of the operating system depends on your technology stack. Common choices include Linux distributions (such as Ubuntu or CentOS) for server environments.

2. Web Server:

- A web server (e.g., Apache, Nginx) to handle HTTP requests and serve the web application.

3. Database Management System (DBMS):

- A database management system like MySQL, PostgreSQL, or MongoDB to store and retrieve data.

CHAPTER 5 – EXPERIMENTAL RESULTS

1. Define the Scope and Goals:

- The scope of the code review platform encompasses facilitating collaborative and efficient evaluation of software code by providing a centralized environment for developers to submit, review, and discuss code changes.
- The primary goals include improving code quality, identifying and resolving issues early in the development process, and fostering a culture of knowledge sharing and collaboration among development teams.

2. Collect and Develop Sign Language Databases:

- Develop a mechanism to store snapshots of code changes associated with each review, allowing for historical tracking and comparison of code versions over time.
- Implement a database schema to capture metadata associated with code reviews.

PROGRAM:

IN PYTHON

```
import tkinter as tk
from tkinter import scrolledtext

def correct_syntax_errors(code):
    corrected_code = ""
    lines = code.split("\n")
    current_indentation = 0

    for line in lines:
        trimmed_line = line.strip()

        # Adjust indentation based on braces
        if trimmed_line.endswith("{"):
            current_indentation += 1

        # Preserve existing indentation
        corrected_code += " " * current_indentation

        # Correct syntax errors and append the line
        if trimmed_line:
            if trimmed_line.startswith("}"):
                current_indentation = max(current_indentation - 1, 0)

            if not trimmed_line.endswith(";") and not trimmed_line.endswith("{") and not
trimmed_line.endswith("}"):
                corrected_code += trimmed_line + ";"
            else:
```

```

        corrected_code += trimmed_line
    else:
        corrected_code += trimmed_line

    corrected_code += "\n"

return corrected_code

def correct_code():
    code_content = input_text.get("1.0",'end-1c')
    corrected_code = correct_syntax_errors(code_content)
    output_text.delete("1.0", tk.END)
    output_text.insert(tk.END, corrected_code)

# Create the main window
root = tk.Tk()
root.title("Code Review GUI")

# Input Textbox
input_text = scrolledtext.ScrolledText(root, width=40, height=10)
input_text.pack(padx=10, pady=10)

# Button to Correct Code
correct_button = tk.Button(root, text="Correct Code", command=correct_code)
correct_button.pack(pady=5)

# Output Textbox
output_text = scrolledtext.ScrolledText(root, width=40, height=10)
output_text.pack(padx=10, pady=10)

# Run the GUI
root.mainloop()

```

IN JAVA

```
import java.util.Scanner;

public class SyntaxErrorDetector {

    public static String correctSyntaxErrors(String code) {
        StringBuilder correctedCode = new StringBuilder();
        Scanner scanner = new Scanner(code);
        int currentIndentation = 0;

        while (scanner.hasNextLine()) {
            String line = scanner.nextLine().trim();

            // Adjust indentation based on braces
            if (line.endsWith("{")) {
                currentIndentation++;
            }

            // Preserve existing indentation
            for (int i = 0; i < currentIndentation; i++) {
                correctedCode.append("  ");
            }

            // Correct syntax errors and append the line
            if (!line.isEmpty()) {
                if (line.contains("{")) {
                    currentIndentation++;
                }

                if (currentIndentation > 0 && line.contains("}")) {
                    currentIndentation--;
                }
            }
        }
    }
}
```

```

        if (!line.endsWith(";") && !line.endsWith("{") && !line.endsWith("}")) {
            correctedCode.append(line).append(";");
        } else {
            correctedCode.append(line);
        }
    } else {
        correctedCode.append(line);
    }

    correctedCode.append("\n");
}
scanner.close();
return correctedCode.toString();
}

public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);
    System.out.println("Enter the code for review (press Enter twice to finish:");
    StringBuilder codeContent = new StringBuilder();
    while (true) {
        String line = scanner.nextLine();
        if (line.isEmpty()) {
            break;
        }
        codeContent.append(line).append("\n");
    }

    // Correct the syntax errors while maintaining indentation
    String correctedCode = correctSyntaxErrors(codeContent.toString());
    // Display corrected code
    System.out.println("\nCorrected Code:\n" + correctedCode);
    scanner.close();
}
}

```



```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains
Enter the code for review (press Enter twice to finish):
public class SampleCode {
    public static void main(String[] args) {
        int x = 5
        if (x > 0) {
            System.out.println("Positive")
        }
    }
}

Corrected Code:
public class SampleCode {
    public static void main(String[] args) {
        int x = 5;
        if (x > 0) {
            System.out.println("Positive");
        }
    }
}

Process finished with exit code 0
|
```

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\J
Enter the code for review (press Enter twice to finish):
public class SampleCode {
    public static void main(String[] args) {
        int x = 5
        if (x > 0) {
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Corrected Code:
public class SampleCode {
    public static void main(String[] args) {
        int x = 5;
        if (x > 0) {
            System.out.println("Positive");
        }
    }
}

Process finished with exit code 0
```

Code Review GUI

```
public class SampleCode {
    public static void main(String[] arg
s) {
        System.out.println("Hello");
    }
}
```

Correct Code

```
public class SampleCode {
    public static void main(String[]
args) {
        System.out.println("Hello");
    }
}
```

Code Review GUI

```
public class SampleCode {
    public static void main(String[] arg
s) {
        int x = 5
        if (x > 0) {
            System.out.println("Positive
")|
        }
    }
}
```

Correct Code

```
public class SampleCode {
    public static void main(String[]
args) {
        int x = 5;
        if (x > 0) {
            System.out.println("Positive
");
        }
    }
}
```

CHAPTER 6 - CONCLUSION

In conclusion, the development of a robust code review platform is a pivotal step in fostering a collaborative and quality-driven software development environment. By providing a structured framework for code evaluations, the platform aims to enhance code quality, accelerate issue identification and resolution, and cultivate a culture of continuous improvement within development teams. The implementation involves meticulous consideration of hardware and software requirements, coupled with a comprehensive database design to capture essential metadata and code snapshots. As the code review platform evolves, it stands as a cornerstone in promoting effective collaboration, knowledge sharing, and overall excellence in the software development lifecycle.

As the code review platform takes shape, its significance extends beyond the technical realm, influencing the broader aspects of team dynamics and project outcomes. By streamlining the review process, the platform not only contributes to code quality but also nurtures a collaborative atmosphere where developers actively engage in constructive discussions, share insights, and collectively drive the project towards success. The continuous evolution of the platform, guided by user feedback and emerging industry practices, underscores its adaptability and commitment to facilitating efficient, transparent, and impactful code reviews that are fundamental to the success of any software development endeavor.

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